graphify

Complete Guide: Multi-Agent Context Switching

Claude Code · Cursor · Windsurf · Antigravity · Codex · Aider · Copilot

71.5×
fewer tokens per query

1 command
to switch agents

0 tokens
wasted on re-reading

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01 — Overview: Graphify kya hai?

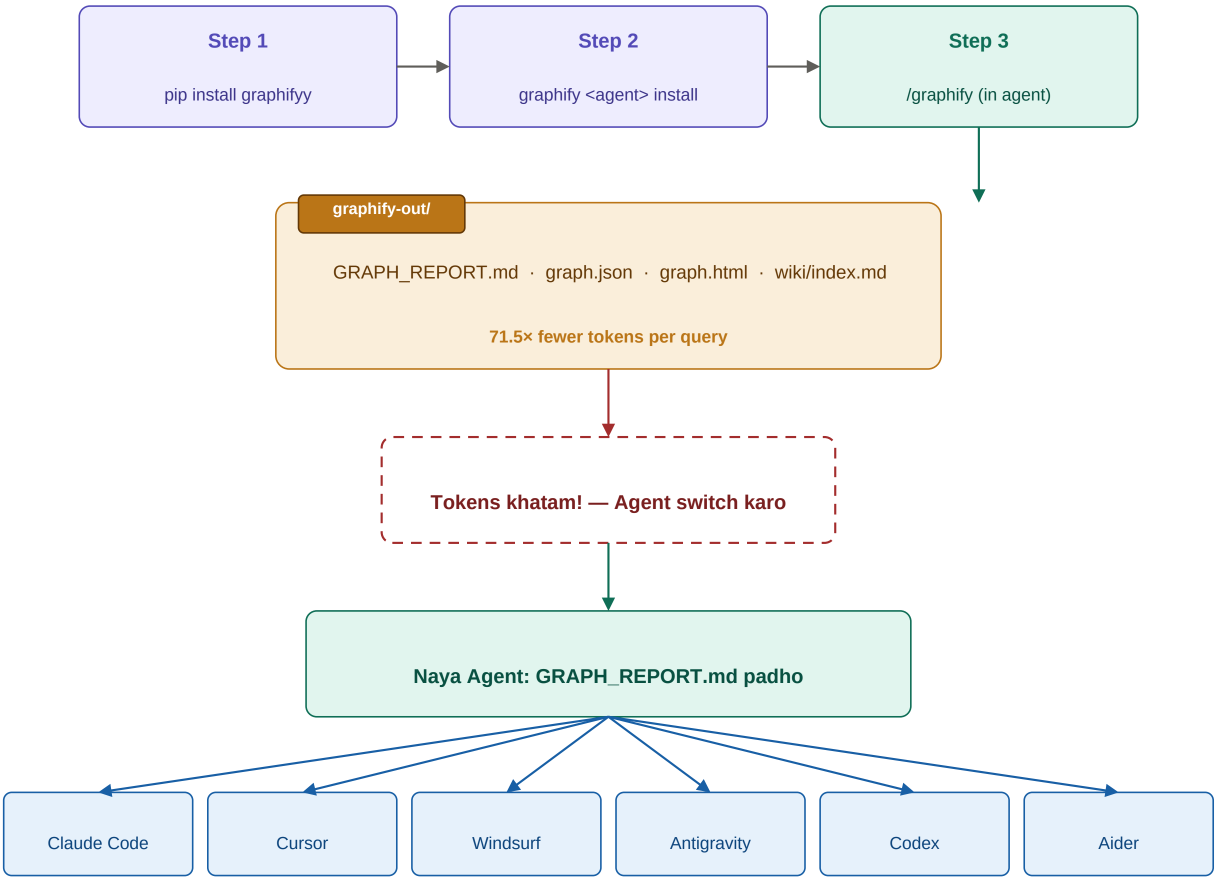
Jab bhi tum ek AI coding agent (jaise Claude Code, Cursor ya Windsurf) se koi bada codebase samjhne ko kehte ho — woh har baar raw files scratch se padhta hai. 50-file project mein yeh lakhon tokens burn kar deta hai — aur phir bhi cross-file connections miss ho jaate hain.

Graphify is solve karta hai. Yeh ek baar codebase ka analysis karta hai aur ek **persistent knowledge graph** banata hai — jo disk pe save rehta hai. Agla query raw files nahi padhta, sirf compact graph padhta hai. Result: **71.5x fewer tokens per query.**

Bina Graphify (Problem)	Graphify ke saath (Solution)
Har query mein poori codebase dobara padhi jaati hai	Ek baar graph bano, baaki queries graph se answer hoti hain
Agent switch karne par saara context lost ho jaata hai	graphify-out/ folder context preserve karta hai
Lakhon tokens waste, slow responses	71.5x fewer tokens, instant context
Cross-file connections miss ho jaate hain	Full dependency graph, call graphs, rationale sab capture
Har agent apne session mein sab bhool jaata hai	SHA256 cache — sirf changed files reprocess hoti hain

Graphify open-source hai (**github.com/safishamsi/graphify**) aur April 2026 release ke baad 24,000+ GitHub stars haasil kar chuka hai. Yeh Claude Code, Cursor, Windsurf, Antigravity, Codex, Aider, GitHub Copilot CLI, aur aur bhi bahut saare agents ko support karta hai.

02 — Workflow Diagram



Har agent graphify-out/ se codebase instantly samjhta hai — bina raw files padhe

Upar diagram mein poora flow dikhaya gaya hai: Install → Agent mein setup → Graph banao → Tokens khatam hone par switch karo → Naya agent graph padh ke instantly ready ho jaata hai.

03 — Step 1: Installation

Graphify ek Python package hai. Ek baar install karo, saare projects mein kaam aayega:

```
pip install graphifyy
```

> Note: Package name 'graphifyy' hai (double-y) — PyPI pe yahi naam available tha.

Prerequisites:

- **Python 3.9+** — python --version se check karo
- **Anthropic API Key** — LLM extraction ke liye (docs/PDFs/images ke liye)
- **Git (optional)** — Auto-sync hooks ke liye

04 — Step 2: Agent mein Skill Install karo

Graphify ko apne coding agent ke andar /graphify command ke roop mein register karna hota hai. Har agent ka alag install command hai:

Claude Code	`graphify claude install`
Cursor	`graphify cursor install`
Windsurf	`graphify windsurf install`
Google Antigravity	`graphify antigravity install`
GitHub Copilot CLI	`graphify copilot install`
VS Code (Copilot Chat)	`graphify vscode install`
OpenAI Codex	`graphify codex install`
Aider	`graphify aider install`

> Yeh command project root mein run karo. Yeh SKILL.md, hooks, ya rules file install karta hai — agent ke hisaab se alag-alag

05 — Step 3: Pehli Baar Graph Banao

Project root mein jaao aur agent ke chat mein type karo:

```
/graphify
```

Graphify ek **7-stage pipeline** run karta hai:

`detect()`	Directory scan karta hai — 25 programming languages + docs + PDFs + images + video
`extract()`	3-pass extraction: AST (code) + Whisper (audio/video) + Claude subagents (docs)
`build_graph()`	NetworkX graph mein merge karta hai — nodes: concepts/classes/functions
`cluster()`	Leiden community detection se related components group karta hai
`analyze()`	'God nodes' dhundta hai — high-degree concepts jo sab se connect hain
`report()`	GRAPH_REPORT.md generate karta hai — plain language summary
`export()`	graph.json, graph.html, wiki/, MCP server — multiple formats

Output — graphify-out/ folder mein kya milta hai:

`GRAPH_REPORT.md`	■ **Sabse important**	Plain language mein poora architecture summary. Naye agent ko sirf yahi file dikhao.
`graph.json`	**Core data**	Machine-readable complete graph. 71.5x token reduction ka raaz yahi hai.
`graph.html`	**Visual**	Browser mein interactive visualization. Nodes click karo, communities explore karo.
`wiki/index.md`	**Navigation**	Har community ke liye Wikipedia-style articles. Agent yahan se navigate karta hai.
`transcripts/`	**Cache**	Video/audio transcriptions cache — dobara process nahi hogi.

06 — Step 4: Token Khatam Hone Par Agent Switch

Yeh is poori system ki main value hai. Jab ek agent ke tokens khatam ho jaayein:

4a. Pehle (agar possible ho) — purane agent mein progress note karo:

```
graphify-out/GRAPH_REPORT.md mein current task aur progress
update karo. Main [TASK_NAME] par kaam kar raha tha,
[FILE_NAME] mein [FUNCTION] implement karna baaki hai.
```

4b. Naye agent mein — yeh pehla message do:

```
graphify-out/GRAPH_REPORT.md padho — yeh ek knowledge graph
hai jo is codebase ka poora structure describe karta hai.

Phir graph.json se specific architecture detail samjho.

Ab mujhe [NEXT_TASK] karna hai. Shuru karo.
```

> Bus itna! Naya agent bina ek bhi raw file padhe poori codebase ki structure samajh jaata hai. SHA256 cache ensure karta ha

Agent switch karne ka checklist:

Purana agent	Tokens khatam hone se pehle GRAPH_REPORT.md mein progress note karo
Naya agent	/graphify install check karo — ya manually skill path set karo
Pehla message	GRAPH_REPORT.md padhne ko kaho, phir task do
Verify	Agent se puchho: 'Codebase ki main architecture kya hai?' — sahi jawab aana chahiye

07 — Step 5: Graph Auto-Update Rakhna

Codebase badalta rehta hai — graph ko fresh rakhne ke teen tarike hain:

Git Hook (Recommended) `graphify hook install`

Har git commit ke baad graph automatically rebuild hota hai. Branch switch karne par bhi. Background process nahi chahiye.

Watch Mode (Background) `graphify --watch`

Background terminal mein chalao. Code files save hone par instant rebuild. Docs/images ke liye notification aata hai — manual --update run karo.

Manual Update `graphify --update`

Jab bhi chahho manually run karo. SHA256 cache ensure karta hai ki sirf changed files reprocess hoti hain — time waste nahi hota.

.graphifyignore — Kya Skip Karein

Project root mein .graphifyignore banao — same syntax as .gitignore:

```
node_modules/
dist/
build/
.env
*.log
__pycache__/
.git/
```

08 — Agent-wise Command Reference

Agent	Install	Trigger
Claude Code	`graphify claude install`	`/graphify`
Cursor	`graphify cursor install`	`/graphify`
Windsurf	`graphify windsurf install`	`/graphify`
Google Antigravity	`graphify antigravity install`	`/graphify`
GitHub Copilot CLI	`graphify copilot install`	`/graphify`
VS Code	`graphify vscode install`	`chat mein type: /graphify`
Aider	`graphify aider install`	`/graphify`

Always-on vs Explicit Trigger — kya fark hai?

Always-on: Har session start par automatically graph context inject hota hai (Claude Code, VS Code, Antigravity).

Explicit trigger: Tum /graphify type karo tab graph banata/padhta hai (Cursor, Windsurf, Aider).

MCP Server Mode

Agar agent MCP (Model Context Protocol) support karta hai — jaise Claude Code — to graph ko MCP server ke roop mein serve karo. Agent structured tools se query karta hai, text paste nahi karna padta:

```
python -m graphify.serve graphify-out/graph.json

# Available MCP tools:
# query_graph, get_node, get_neighbors, shortest_path
```

10 — Pro Tips & Best Practices

Wiki Mode use karo

```
graphify --wiki
```

Wiki mode mein Wikipedia-style articles banate hain — ek per community. Agent index.md se navigate karta hai — JSON parse karne se zyada natural. Best for agents jo natural language prefer karte hain.

Multimodal input dalo

```
# Koi bhi file drop karo project mein
```

Graphify sirf code nahi padhta — PDFs, architecture diagrams, whiteboard photos, screen recordings, design docs — sab kuch. Whisper se audio/video transcribe hoti hai. Jitna zyada context doge, graph utna rich hoga.

Obsidian Vault export karo

```
graphify --obsidian
```

Obsidian-compatible vault generate hota hai — visual graph view mein explore karo. Team ke saath share karne ke liye best option hai.

God nodes ko samjho

```
# GRAPH_REPORT.md mein 'God Nodes' section dekho
```

God nodes woh concepts hain jinse sabse zyada cheezein connect hoti hain. Koi bhi feature banana ho — pehle god nodes dekho, wahan se shuru karo. Yeh codebase ki 'spine' hoti hai.

graphify-out/ ko .gitignore mein mat daalo

```
# graphify-out/ ko git mein commit karo!
```

Team ke saare members aur saare agents same graph share karein. graph.json commit karo taaki CI mein bhi use ho sake. graph.html ko team ke wiki ya Notion mein share karo.

11 — Quick Reference Table

Task	Command	Notes
Install karo	`pip install graphifyy`	Python 3.9+ required
Claude Code mein setup	`graphify claude install`	Hook install karta hai
Cursor mein setup	`graphify cursor install`	.cursor/rules/ mein SKILL.md
Windsurf mein setup	`graphify windsurf install`	.windsurf/rules/ mein SKILL.md
Antigravity mein setup	`graphify antigravity install`	Always-on rules
Graph banao	`/graphify (agent mein)`	Pehli baar 2-5 min lag sakte hain
Graph update karo	`graphify --update`	Sirf changed files process hoti hain
Auto-watch mode	`graphify --watch`	Background terminal mein chalao
Git hook install	`graphify hook install`	Post-commit auto rebuild
Wiki mode	`graphify --wiki`	Agent-crawlable articles
Obsidian vault	`graphify --obsidian`	Visual graph exploration
MCP server	`python -m graphify.serve graphify-out/graph.json`	Structured graph access
Naya agent onboard	`GRAPH_REPORT.md padho pehle`	Phir task do
Uninstall	`graphify uninstall`	Reverse of install

Resources

GitHub Repository: github.com/safishamsi/graphify

PyPI Package: pypi.org/project/graphifyy

Documentation: graphify.net

Knowledge Graph Guide: graphify.net/knowledge-graph-for-ai-coding-assistants.html